

COMSM0127 Immersive Interaction and Audio Design

Week 3, Lab B: Sound Design and Dynamic Audio

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Today's Lab

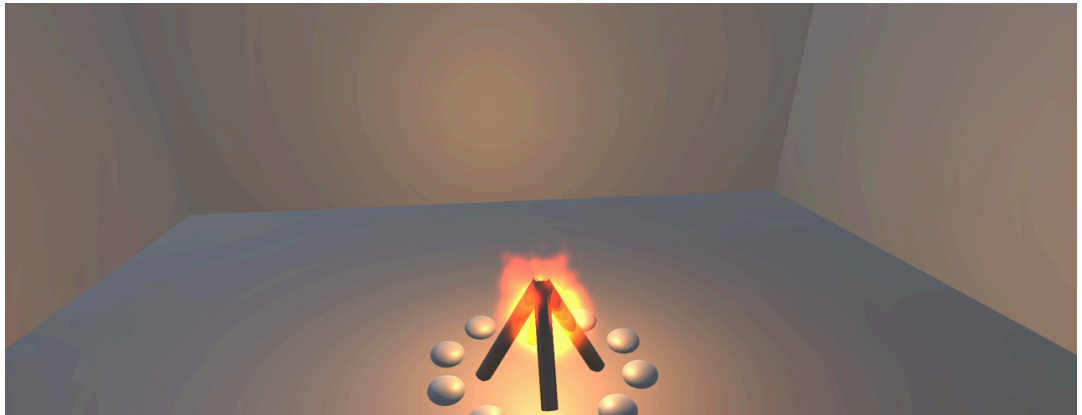
Last week you put a spatialised sound in a room. Today you **make** a sound, and make it change.

You heard the mystery sound at the start. Rebuild it two ways:

- **Part A, Recorded:** record it, trim it, and play it back so it never repeats.
- **Part B, Procedural:** sculpt it from noise in code, and make it respond to the game.

	09:22 – 09:50	**09:56 – 10:22**
Groups 1, 2	**Part A: Recorded**	Part B: Procedural
Groups 3, 4	**Part B: Procedural**	Part A: Recorded

At **09:50** everyone stops for six minutes before swapping. At **10:25** everyone comes back together to listen.



Goals to Complete Before the End of the Lab

By the end of this lab you should have:

1. Listened to the mystery sound and named the parts it is made of.
2. **Part A:** recorded those parts and played them back so no two plays are the same.
3. **Part B:** sculpted those parts from noise and made one respond to the fire's Intensity.
4. Added **one line of C#** in each part.
5. Played your fire to the room.

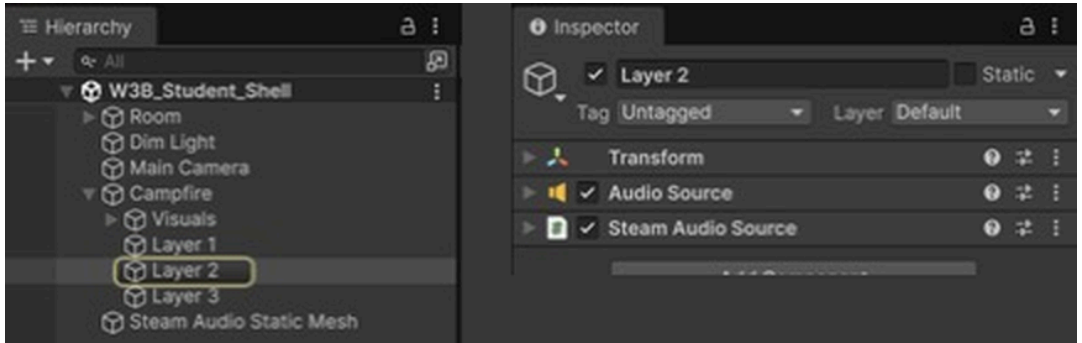
Basic goal: both parts working on the campfire.
Stretch goal: one fire with recorded and procedural layers together.

Opening the project

1. Download the project: **github.com/RichSF1210/W3B_FireLab**, green **Code** button, **Download ZIP**. Unzip it to a short path, e.g. `C:\Unity\W3B_FireLab`.
2. Unity Hub: **Add > Add project from disk**, choose the folder, open in **6000.3.19f1**. The first open takes a few minutes.
3. Project window: `Assets/W3B_Fire/Scenes`, double-click **W3B_Student_Shell**. **This is the only scene you need.**
4. Press Play. Walk with **W A S D**; hold the right mouse button to look.

The room, Steam Audio and the campfire are done. The campfire has three empty layers: **Layer 1, 2, 3**, each with a spatialised Audio Source, as in Week 2 Lab B.

The fire is silent. That is your job.



First: listen

Headphones on, eyes closed. Listen to the mystery sound again: `Assets/W3B_Fire/Audio/Mystery`.

As a group, agree:

- How many separate parts can you hear?
- Describe each one: high or low? Constant, or in short bursts?
- Which part tells you most strongly what the sound is?

Give each part a name of your own and write it down. **One part per layer.**

A

Part A: Recorded

A1. Record (about 10 minutes)

Take a Zoom recorder outside the lab. For each part on your list, find something that makes a similar sound: scrunch, snap, spray, blow, rub, tear.

- Get close. Keep it quiet around you.
- Record **several takes** of each part. For short sounds, aim for **at least four** different ones.
- About 10 seconds per take is plenty. Say the part's name before each take, so you can find it later.

Safety: do not record people without asking.

A2. Get the files off and tidy them (about 10 minutes)

1. **Off the recorder:** plug the Zoom into a lab PC with its USB cable (choose *Card Reader* on its screen), or take out the SD card. Copy the WAV files to your desktop.
2. **Audacity:** open a file. Select the part you want, **Edit > Remove Special > Trim Audio** (or Ctrl+T).
3. **Short sounds:** one sound per file, starting right on the sound.
4. **Constant sounds:** a few seconds that loop without a click (cut where the waveform crosses the centre line).
5. **File > Export Audio**, as **WAV**. Name it clearly, with no spaces: `snap_01.wav`, `snap_02.wav`, `air_loop.wav`.

Drag the WAVs into Unity's Project window, into `Assets/MyAudio`. Unity accepts WAV, AIFF, MP3 and OGG. **The names do not matter to the scripts.**

A3. Play them back

Constant parts (a loop): select a layer, **Add Component > SampleLoopLayer**, drag your loop into the Audio Source's **AudioClip**.

Short parts (several files): select a layer, **Add Component > MyRandomPlayer**. Lock the Inspector (padlock, top right), drag your files onto **Clips**, then unlock.

Press Play and listen. Then listen again. **What gives it away as a recording?**



A4. Your one line: make it vary

Double-click the **Script** field on My Random Player. Find `TODO`. The line is already there: **delete the two slashes** in front of it.

```
source.pitch = Random.Range(0.85f, 1.2f);
```

`Random.Range(0.85f, 1.2f)` picks a random number between 0.85 and 1.2. 1.0 is normal pitch; lower is deeper and slower, higher is brighter and faster.

Save (Ctrl+S). Wait for Unity's spinner. Press Play.

The files never change. The playback never repeats. This is dynamic audio from recordings.

```
void Update()
{
    if (clips == null || clips.Length == 0 || Time.time < nextTime) return;

    AudioClip clip = clips[Random.Range(0, clips.Length)];

    // TODO: YOUR ONE LINE. Delete the two slashes.
    source.pitch = Random.Range(0.85f, 1.2f);

    source.PlayOneShot(clip);
    nextTime = Time.time + Random.Range(shortestGap, longestGap);
}
```

B

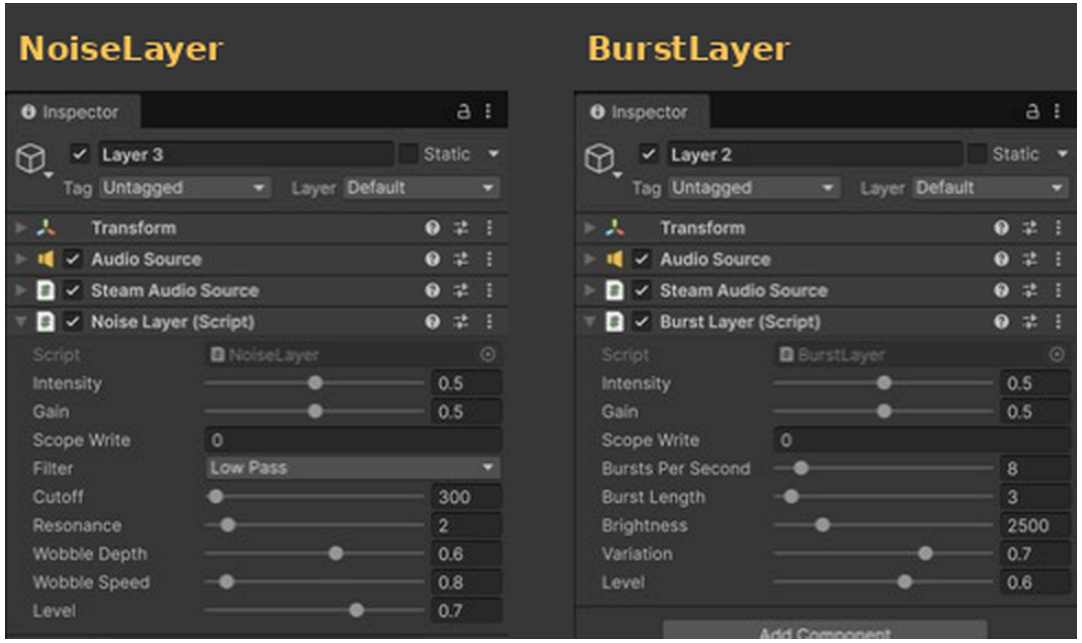
Part B: Procedural

B1. Two building blocks

No audio files: these make sound from numbers, as it plays.

- **NoiseLayer:** noise through a filter, with a slow wobble. For a **constant** part. Sliders: filter type, cutoff, wobble, level.
- **BurstLayer:** very short bursts of noise at random moments. For a part made of **separate events**. Sliders: bursts per second, length, brightness, variation, level.

Select a layer, **Add Component**, and add one. Each script explains every line inside it.



B2. Sculpt by ear

Press Play. Move the sliders **while you listen**.

- NoiseLayer, Low Pass, low cutoff: what does it remind you of? High Pass, high cutoff?
- BurstLayer, fast, short and bright: what is it? Slow, long and dark?

Match each layer to one part on your list. Set the other layers' **Level** to 0 while you tune one.

Sliders changed in Play mode reset when you stop. Write your best values down, or right-click the component: *Copy Component*, stop, then *Paste Component Values*.

B3. Your one line: make it respond

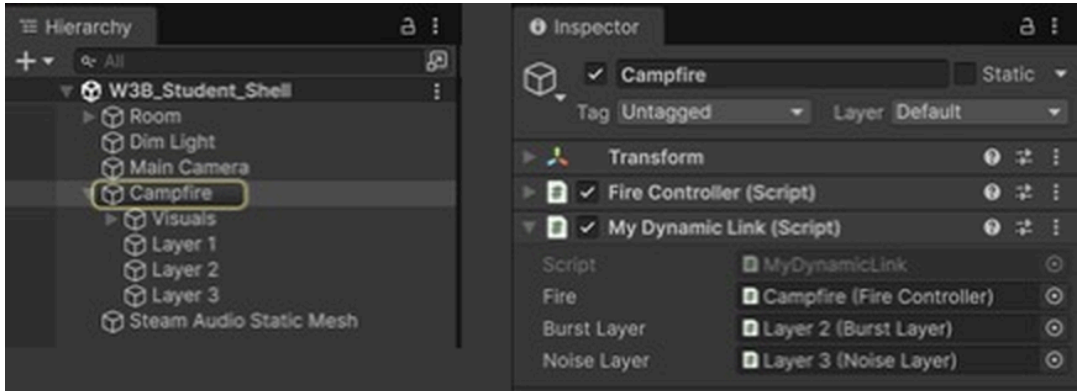
The Campfire has a **FireController** with an **Intensity** slider: 0 is embers, 1 is a blaze.

1. Select **Campfire**, **Add Component > MyDynamicLink**.
2. Drag **Campfire** into **Fire**, and your layers into **Burst Layer** and **Noise Layer**.
3. Double-click its Script field. Under `TODO`, **delete the two slashes** in front of one line:

```
burstLayer.burstsPerSecond = Mathf.Lerp(1f, 40f, i);
```

`Mathf.Lerp(1, 40, i)` goes from 1 to 40 as `i`, the fire's Intensity, goes from 0 to 1. This is **RTPC**, real-time parameter control: a value from the game drives a parameter of the sound.

Save, press Play, select Campfire and drag **Intensity**. Is it more than volume?



Stuck?

- **No sound?** Is the layer's Level above 0? The Audio Source volume up? The object ticked?
- **Red error in the Console?** Double-click it: it opens the script at the problem. Check the line ends with `;` and the spelling matches exactly.
- **Changes vanished?** You made them while playing. Make them again when stopped.
- **Recorder trouble, or out of time?** Use the recordings in `Assets/W3B_Fire/Audio/Fire_Recorded`.
- **Too loud?** Start with headphones low: bursts can be sudden.

Ask us. That is what we are here for.



Stretch

Stretch tasks

- **Combine:** one fire with a recorded layer and a procedural layer. Which parts work better each way?
- **Interact:** add `AddLogOnKey` to the Campfire and press **L** to throw on a log. In your exhibit, call `AddLog` from a grab instead.
- **Adapt:** on FireController, tick **Burn Down**.
- **More lines:** use both lines in MyDynamicLink, or change the numbers. Can you make the fire sound bigger, not just louder?

At 10:25, be ready to tell us

Bring one fire to play through the speaker, and tell us:

1. What parts did you hear in the mystery sound? What did you call them?
2. Which part worked better recorded, and which procedural?
3. What did each of your lines of code change?

For your coursework: which sound in your exhibits needs to be *dynamic*? Bring it to Monday's session.

Thank you